Computer Methods and Programs in Biomedicine Update 3 (2023) 100106

Contents lists available at ScienceDirect

Computer Methods and Programs in Biomedicine Update

journal homepage: www.sciencedirect.com/journal/computer-methods-and-programs-in-biomedicine-update

Machine and deep learning identified metabolites and clinical features associated with gallstone disease

Nourah M Salem [a], Khadijah M Jack [a,*], Haiwei Gu [c], Ashok Kumar [b], Marlene Garcia [a], Ping Yang [b], Valentin Dinu [a]

[a] College of Health Solutions, Arizona State University, Arizona, 85004, United States of America
[b] Department of Quantitative Health Science, Mayo Clinic, Arizona, 85259, United States of America
[c] Florida International University, Florida, 33199, United States of America

ARTICLE INFO

Keywords:
Machine learning
Feature selection
Gallstone disease
Sex disparities
Metabolites

ABSTRACT

Machine Learning (ML) algorithms can be used to analyze metabolomic expression data to explore the association between metabolite expression and disease etiology. In this study, we used and compared the performance of ML algorithms to analyze polar aqueous and blood-based lipid-based metabolites to identify meaningful patterns correlated with the development of gallstone disease (GSD) while examining the sex disparity. We also developed ML approaches that used clinical risk factors for predicting GSD, including age, obesity, body mass index, hemoglobin A1c, dyslipidemia index cholesterol to high-density lipoprotein ratio (CHOL/HDL). A more powerful data fusion model that combines both metabolomic and clinical features achieved accuracy of 83% for accurate prediction of the presence of GSD.

Introduction

Gallstones or Cholelithiasis, a disease characterized by cholesterol or pigment stones located in the gallbladder or bile ducts, is the most common risk factor for the development of gallbladder cancer (GbCa); and gallstones occur in up to 80% of GbCa patients [1,2]. GbCa is the most frequent malignancy of the biliary tract. GbCa is sixth in overall abdominal cancer incidence and a lethal malignancy with median survival time of about six months. Studies have shown several regions of the globe have an increased prevalence of GbCa including Korea, Japan, North India, and New Mexico [1]. Multiple modifiable and unmodifiable risk factors are known to contribute to the development of GSD including female sex, diet, vitamin deficiencies, toxic ingestions, aflatoxins, socioeconomic status, reduced physical activity, and race/-ethnicity. In the U.S., higher incidence in Native American and Hispanic people compared to non-Hispanic Whites.

Metabolites in Biomedical Research: Metabolites, small molecules with the molecular weight lower than 1500 Dalton, play an important role in cell survival and normal growth including mental and physical development. Therefore, their production level in the body can represent the biological state of an organism [3]. In this study, we used machine learning algorithms to evaluate the association between metabolites and gallstone disease onset.

Machine Learning (ML) in Metabolomic Studies: ML is a growing area of investigation in the field of artificial intelligence with a main principle of enabling machines/computers to learn from data to make decisions. In biomedicine, ML allows scientists to integrate large scale heterogeneous data to better understand the basis of disease for the purpose of identifying the disease in early stages and developing optimal therapeutic approach [4].

We have previously developed methods for combining machine learning and statistical approaches to analyze the association between -omics data and disease [5], and we have applied these methods to study the etiology of various disorders [6–16]. ML algorithms in the context of this study include Support vector machine (SVM), Random forests (RF), and Artificial neural networks (ANNs) [17]. SVM is a supervised learning algorithms used for classification and regression that identifies the line or hyperplane for an N-dimensional space (where N is the number of classes) with the maximum margin (the maximum distance between data points of classes), for future data points to be classified with more confidence [18,19]. ANN algorithms are used for classification, prediction, and pattern recognition. A neural network is arranged in a similar manner to the human brain. It consists of neurons (nodes) connected in input and output layer layers. The first layer, input layer,

* Corresponding author.
E-mail address: kcrosby2@asu.edu (K.M. Jack).

https://doi.org/10.1016/j.cmpbup.2023.100106

Available online 3 May 2023
2666-9900/© 2023 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

receives various forms and structures of data. The hidden layers make up the body of the network and mathematically find a discriminating pattern for the features. The last layer, the output layer, produces the predicted decision. ANNs can learn and construct non-linear models for complex relationships [20]. Deep Learning (DL) are ML approaches that include Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Transformers and can be used for more complex relationships [21]. Transformers can be used for more complex relationships, in which it can extract its contributing features from the data [21].

In this study we used and compared ML and DL approaches to classify the risk of GSD, using as predictors metabolomic expression levels in blood and clinical risk factors: age, obesity, body mass index, hemoglobin A1c, dyslipidemia index cholesterol to high-density lipoprotein ratio (CHOL/HDL). The models were also used to evaluate the association of sex (male vs. female) with gallstone presence. We assessed the performance of various ML algorithms to identify the best performing models using molecular and clinical data, separately and combined using data fusion approaches.

Methods

Samples

To identify candidate metabolites contributing to GbCa development, we have chosen a matched case-control design to efficiently adjust for confounding effects of known risk factors of GbCa as covariates; they are high BMI (≥30), A1c (≥ 5.7), and dyslipidemia. Table 1 displays the characteristics of the sample matching results. The cases and controls were matched based on the available study subjects in the Sangre Por Salud Biobank. In brief, we ascertained all cases ($n = 100$ [by chance not by choice]) and the best matched controls to the cases; matching was done 1:1 in all categories accompanied with one of more comorbid disease (covariates). Six of the 100 cases did not have any of the three comorbid diseases, along with a higher number of matched healthy controls, serving as the baseline for improved power. Overall, cases and controls were matched by age (± 5 years) and sex. Of note, 6 in "healthy" ("no other disease") group with Gb/Gs history (5 women, 1 man) were matched to 24 in "healthy" group without Gb/Gs history (20 women, 4 men), a total of 30 to increase the baseline measures and 94 in other 7 pre-defined groups with Gb/Gs history matched 1:1 in "healthy" group without Gb/Gs history (total of 188 subjects).

Data collection

We analyzed 218 plasma samples of 100 GSD patients (cases labeled as 1) and 118 healthy controls (labeled as 0). The data was de-identified to remove all patient identification information and attained electronically from Sangre Por Salud (SPS) Biobank and Mountain Park Health Center (MPHC), which also provided plasma samples. There were 181 females with mean age of 49.8 (SD 7.9), and 37 males, mean age of 49.2 (SD 6.5). The comorbid conditions under study are known to be risk factors for gallstones including age, obesity, body mass index, hemoglobin A1c, dyslipidemia index cholesterol to high-density lipoprotein ratio (CHOL/HDL). Plasma metabolite profiles were generated from metabolomics profiling methods and platforms including liquid chromatography tandem mass spectrometry (LC-MS/MS), gas chromatography (GC)-MS, and LC—Orbitrap MS. The use of multiple methods allowed for a very wide range of aqueous metabolites and lipids levels to be detected. A total of 302 metabolites were measured including both polar aqueous and lipid-based metabolites such as nucleic acids, sugars, amino acids, and small organic molecules, which are essential for metabolic pathways required for survival, growth, and development.

Table 1
Study sample characteristics (all age ≥ 40). BMI (Body Mass Index): BMI ≥ 30. Serum A1c ≥ 5.7 (prediabetes threshold). Lipid: presence of dyslipidemia. Healthy (no other disease): BMI < 30, A1c < 5.7 and no dyslipidemia.

Group	Gb/Gs	No Gb/Gs
No other diseases	6	24
BMI (≥30)	5	5
Dyslipidemia (Lipid)	10	10
A1c (≥5.7)	11	11
BMI, Lipid	8	8
BMI, A1c	13	13
Lipid, A1c	16	16
BMI, Lipid, A1c	31	31
Total	100	118

Analysis steps and methodology

The analysis was conducted in three key steps (Fig. 1):

1 The pre-processing step, where the data was scaled, standardized, best performing algorithm, and optimal number of features were identified, in 3 sub-steps.
2 The second step, where classification models were developed in 2 sub-steps. The first classification model focused on the association between metabolites and clinical conditions (risk factors previously mentioned) on GSD. The second classification model focused on the association between sex, metabolites, and clinical conditions to detect the effect on sex on GSD.
3 The third step, where a clinical and metabolite joint classification model was developed, using deep machine learning approaches.

Fig. 1 illustrates the analysis pipeline. Below we outline details for each of the analysis steps.

Step 1.1: Data Preprocessing: Analyzing metabolic data can be

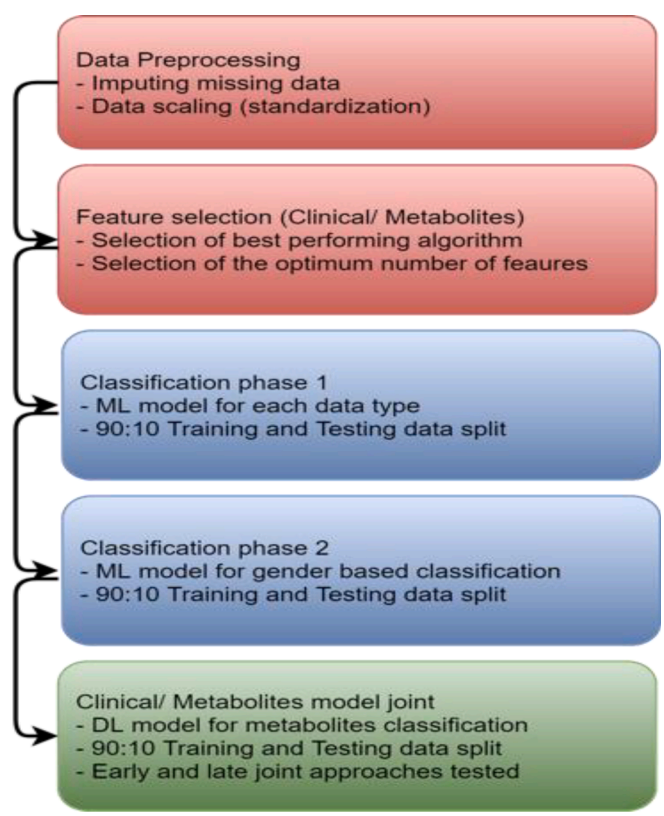

Fig. 1. Clinical and Metabolite Data Analysis Pipeline. The top two red blocks are the data preprocessing and feature and algorithm selection steps, the middle two blue blocks are training/testing two machine learning models, and the bottom green box is a combination of metabolite/clinical data into a single model.

challenging due to missing values generated from inability to detect metabolites in the samples. We addressed the challenge of missing values present in metabolic data by applying the multiple imputation using machine learning to infer the missing value using K-nearest neighbor algorithm. We used the following data standardization equation:

$$z = (x - \mu)/\sigma$$

where μ is the mean (average) and σ is the deviation from the mean to address the variance in values in clinical and metabolite data. Standardization was used instead of normalization since it does not have a bounding range. The data was scaled to increase the accuracy of the classification models (see Results section).

Data imputation was not used in the clinical data analysis since clinical data did not present missing data issues.

Step 1.2: ML Algorithm Selection: The ML models developed were designed to learn from metabolic and clinical data to identify patterns associated with the presence of GSD using a binary classification problem in which cases (GSD present) labeled as 1 and control samples (GSD not present) labeled as 0. Three ML algorithms were trained and tested including SVM, log regression and ANN. The SVM and log regression models were developed using the default architecture from library Sklearn [23]. The ANN model was constructed using a Keras wrapper to build a sequential model with 4 dense layers including the input and output layers. The first layer consisted of 4 nodes, the second layer consisted of 10 nodes, the third layer had 4 nodes, and the output layer had 1 node. After testing multiple node variations to understand the relationship between the number of nodes and model accuracy, we found that the number of layer nodes did not increase the accuracy or the F1 score.

Step 1.3: Feature selection

We trained multiple machine learning algorithms including Random Forest, Extra decision tree, Linear regression, and XGB to sort clinical and metabolites features by importance to avoid dependent features, eliminate noise in samples, increase the model's accuracy and enable faster training [21,22]. The model with the highest F1 score was selected for sorting data according to its contribution to the model learning. Next, we identified the optimal number of metabolites needed for metabolite data training trials by conducting a series of training trials increasing the size to determine the effect of the size on the model accuracy. We performed this process for metabolite and clinical data separately.

Step 2.1: Classification Model Phase 1 - Effect of Clinical Conditions and Metabolites on GSD: Models were trained using a 90:10 model (90% training and 10% testing) using a batch size was 16, number of epochs was set 100, and applied the callback function from Keras (ANN) to stop training before overfitting. Finally, we applied the stratify function to split the test set equally between the two groups.

Step 2.2: Classification Model Phase 2 - Effect of Sex on GSD: The clinical and metabolite data were divided into 4 smaller groups with two female (clinical and metabolite) data sets and two male (clinical and metabolite) data sets. The models were trained and tested separately using a 90:10 model on ANN and GSD as the target class. The same GS binary classification process was used after splitting the data by sex (males labeled as 0 and females labeled as 1). We extracted the top contributing features corresponding to each sex training trial for both the clinical conditions and metabolites. We used the same features used in the first classification model and compared the effect of standardization on the model accuracy and evaluated the F1 score of each ANN model used to understand the ability of the features to predict the GSD.

Step 3: Joint Classification Approaches: A deep learning, convolutional neural network (CNN) model was developed, trained, and tested with clinical and metabolite data. The CNN is a sequential model consisting of three dense layers with the first two layers consisting of 32 nodes while the final layer consisted of one output node. There are two approaches that can be used to achieve this including "early joint" and "late joint" approaches, Fig. 2. The "early joint" approach was a combination of the clinical and metabolite data into a single CSV file and feeding it to the CNN. The "late joint" approach was based on the development of two separate models using CNN for clinical and metabolite data.

Results

Effect of using standardized data on classification accuracy

The scaled and standardized data were used in the metabolite, clinical conditions, sex, and joint classification models. Table 2 highlights the standardization on clinical and metabolite model accuracy. These results suggest that applying data standardization prior to ML algorithms is beneficial (F1 scores in Table 2), and we proceeded with this approach for the ML analysis and results presented in the next analysis steps.

Metabolite feature selection

We identified the optimal number of metabolites needed for metabolite data training trials by conducting a series of training trials increasing the size to determine the effect of the size on the model accuracy. We found that using the top 25 to 50 features had the highest accuracy. Table 3 displays the accuracy for the training trials and F1 score for metabolite data. The highest F1 scores were obtained using Xgboost, with values of 0.87 and 0.86 for class 0 (controls) and class 1 (cases), respectively.

Clinical conditions feature selection

Fig. 3 show the density representation of each clinical feature. Significant features are distributed so the model can learn to discriminate between the GSD cases easier than others. BMI and A1C are discriminated graphically between the two classes, while glucose two-hour and glucose fasting are not shown with discrimination. Redundant features were excluded, and the selected four significant features included: BMI, Hemoglobin A1c, CHOL/HDL ratio and age.

Machine learning algorithm results

Fig. 4 displays the accuracies, precision, recall and F1 scores for the five ML algorithms. ANN and deep learning CNN are the best performing models with accuracy reaching 72% and 82%, respectively.

Effect of sex on GSD

GSD affects disproportionately women, and the underlying disease etiology might be different in women vs. men. We performed separate analysis for GSD in females and males. The summary results are in Table 4. The female clinical model was more accurate than the male model. However, the male metabolite model was more accurate than the female model.

Fig. 5 summarizes the clinical features that were associated most strongly with GSD in males vs. females. Hemoglobin A1c, a diabetes marker, was the top contributing feature in both sexes. The second top contributing feature was BMI in females and Cholesterol/HDL ration in men, which suggest the role of obesity and dyslipidemia in GSD etiology differ in women from men.

Table 5 summarizes the different top metabolites that contributed to GSD case-control classification. Many metabolites are consistent in both sexes; however, the order is different.

Deep learning results using early and late fusion analysis

CNN early and late analysis results were aggregated by calculating

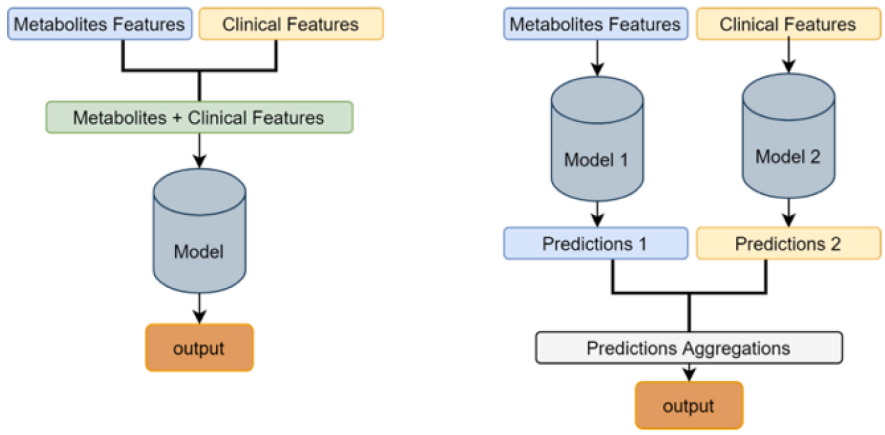

Fig. 2. Joint Classification Models: Early and late model Joint CNN model. Aggregation was applied by taking the average of all predictions from both models in a 10-fold cross validation trial.

Table 2
Standardization on Model Accuracy. Comparison of F1 scores used to assess model performance using standardized vs. non-standardized data. The F1 scores were higher using standardized data.

	Clinical model	Metabolites model
Standardized	0.91	0.82
Non-standardized	0.82	0.59

Table 3
Feature Selection for Metabolite Classification. The left and right F1 values correspond to controls & cases respectively. Xgboost ML had the highest F1 score.

Extra decision tree	Log regression	Xgboost	Random forest
0.67, 0.77	0.50, 0.40	0.87, 0.86	0.62, 0.44

the average predictions of each model as illustrated in Fig. 1 and results are displayed in Table 6. The late joint model that combined classification results after learning separately from metabolomics and clinical data produced overall the most accurate results.

Discussion

Data preprocessing and features selection steps were essential to improving the accuracy of classification ML algorithms. Xgboost algorithm was the best model for our data and the accuracy of training on the metabolites data exceeded 80% compared to the extra decision tree, random forest, and other machine learning selecting features algorithms. Selecting the top features (50 metabolites and 5 clinical conditions) were best for classification learning in the metabolite and clinical conditions models. Neural networks have complex architectures and show the highest accuracies and F1 score when compared to the rest of the machine learning algorithms. When comparing the effect of standardization on data, we found that standardized data was more accurate than unstandardized data. Data standardization and scaling are essential in lowering variance, ensuring a normal distribution can improve the

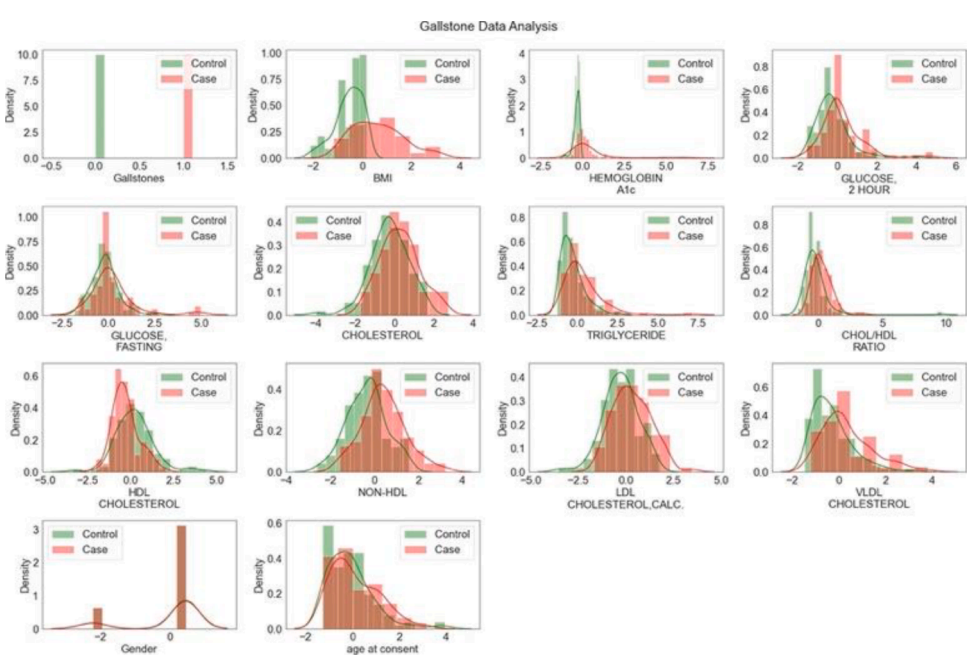

Fig. 3. Density Distribution for clinical features. The differences in distribution for the clinical features for the cases and controls within our data. All redundant features removed, and five significant features were used to train and test the ANN model.

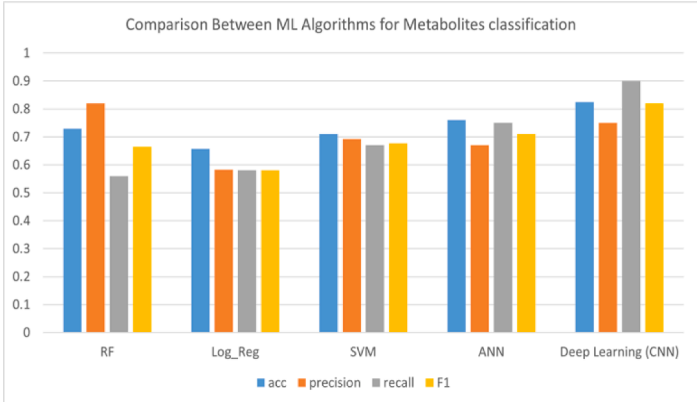

Fig. 4. ML Algorithm Comparison for Metabolite Classification: Random Forest (RF), log regression, Support Vector Machines (SVM), Artificial Neural Network (ANN) and Convolutional Neural Network (CNN). The accuracy, precision, recall, and F1 scores are given for the five ML algorithms. CNN was the best performing model.

Table 4
Sex classification model: F1 scores for the ANN model in the female and male models.

	Females model	Males model
Clinical features	0.84	0.50
Metabolites features	0.63	0.75

learning process for the models resulting in more accurate models. The clinical conditions showed a high accuracy towards predicting GSD. Investigating sex disparities using ML, the data was representative of the GSD prevalence as women tend to have higher prevalence rates of GSD compared to men. BMI was a higher contributing factor in females than in males. In males, HDL and age were stronger contributing factors to GSD than in females. Metabolites contributing to GSD were 100% similar in males and females, however, the order in which they contributed to GSD was different, suggesting potentially different metabolites/pathways might have different levels of influence in GSD in females vs. males.

Biological Validity: Metabolites play an important role in cellular function and the expression of metabolites provide insights into biological functions and identify diagnostic biomarkers for a given disease. The design of the pilot study was aligned with metabolomics workflow experiments. We used an untargeted metabolomics approach to produce plasma metabolite profiles with advanced metabolomic profiling methods including liquid chromatography tandem mass spectrometry (LC-MS/MS), gas chromatography (GC)-MS, and LC—Orbitrap MS to detect various aqueous metabolites and lipids. Next, we developed a robust data analysis pipeline that included data preprocessing, feature and algorithm selection, training and testing models and development of a joint machine learning model. The information produced in the study increase our understanding of the Gb/Gs disease pathology and evaluating treatment options.

Significance: The utilization of machine learning and deep learning for multi-modal data analysis will lead to an enhanced understanding of disease and health [24]. Utilizing machine learning to analyze biological

Table 5
Sex metabolite classification model: The top metabolites for males and females.

Females Contributing Metabolites	Males Contributing Metabolites
Aspartate	4-Imidazoleacetic acid
Leucic acid	Norvaline
Suberic acid	3-Hexenedioic acid
Erythrose	D-Mannose 6-Phosphate
Hydroxyproline	HIAA
Alanine	Alanine
Homoserine/Threonine	Aspartate
HIAA	Lactate
Glucose	Suberic acid
Pipecolinic acid	Leucic acid
Lactate	Homoserine/Threonine
3-Hexenedioic acid	Glucose
Norvaline	Hydroxyproline
D-Mannose 6-Phosphate	Pipecolinic acid
4-Imidazoleacetic acid	Erythrose

Table 6
Joint classification models: The late joint model was more accurate than the early joint model approach.

	Early joint model	Late joint model
Validation accuracy	0.77	0.83

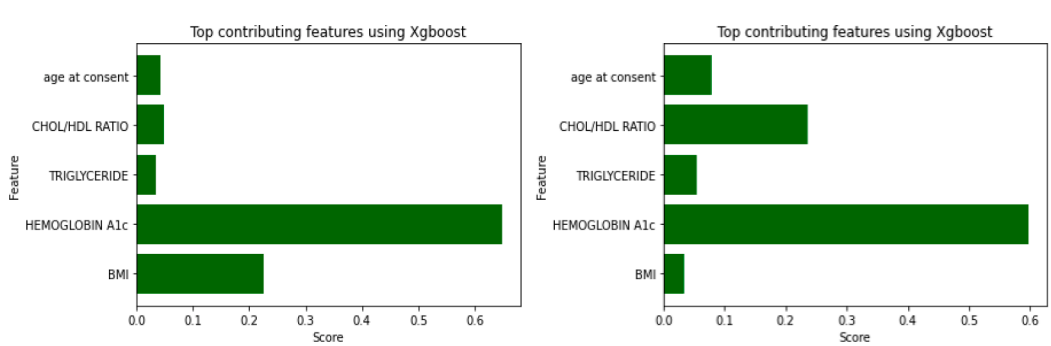

Fig. 5. Sex classification model: The graphs display the clinical features present within each sex that were associated with the onset of GSD. Females are on the left and males are on the right.

data will grow in usage as we move more towards personalized medicine. As the progression to precision medicine continues, it is imperative to develop methods for optimal data analysis and integration of multi-modal data. In this pilot study, machine learning and deep learning algorithms combined to identify which metabolites and clinical risk factors had the strongest association with gallstone disease. This study helps to lay the foundation for utilizing machine learning and deep learning algorithms to identify complex, non-linear patterns within biological datasets.

Limitations of our pilot study: The main limitation was the sample size. Since the study was a pilot study with an expectation to be completed in under 18 months, a smaller and realistic sample size was used. We took considerable attention and extra effect to maximize the power and minimize the bias using a stringently matched case-control design. This study helped to determine the feasibility of future research and research proposals to establish risk factors of gallstone disease, and racial and gender disparities for gallbladder cancer and gallstone disease.

Conclusion

Data mining techniques using machine/deep learning algorithms showed high potential in analyzing metabolomics and clinical data for predicting GSD. In this study, we examined and built standard and deep machine learning models that can classify and predict the potential development of gallstones. We also examined the role of sex on the development of GSD. Our results are support the current literature on GSD that different clinical factors and biological pathways might contribute to GSD in males and in females. Further analysis needs to be done in this area, possibly combining data from more -omics types of data, such as genomics and proteomics, to better understand the etiology of GSD and biological mechanisms. The promise of analytical techniques such as machine learning for this type of analysis remains strong, as our results suggested that deep learning CNN approaches that combined clinical and metabolomic features achieved the highest levels of disease prediction.

Declaration of Competing Interest

The authors do not have a conflict of interest for the study.

References

[1] J.M. Nemunaitis, et al., Gallbladder cancer: review of a rare orphan gastrointestinal cancer with a focus on populations of New Mexico, BMC Cancer 18 (1) (2018) 1–14.
[2] S. Gourgiotis, et al., Gallbladder cancer, The Am. J. Surgery 196 (2) (2008) 252–264.
[3] P. Sen, et al., Deep learning meets metabolomics: a methodological perspective, Brief. Bioinformatics 22 (2) (2021) 1531–1542.
[4] S.-.C. Wang, Artificial Neural Network, in Interdisciplinary Computing in Java Programming, Springer, 2003, pp. 81–100.
[5] V. Dinu, H. Zhao, P.L. Miller, Integrating domain knowledge with statistical and data mining methods for high-density genomic SNP disease association analysis, J. Biomed. Inform. 40 (6) (2007) 750–760.
[6] H.S. Lancaster, et al., Identifying interactive biological pathways associated with reading disability, Brain Behav. 10 (8) (2020) e01735.
[7] B. Peter, et al., Exome sequencing of two siblings with sporadic autism spectrum disorder and severe speech sound disorder suggests pleiotropic and complex effects, Behav. Genet. 49 (4) (2019) 399–414.
[8] B. Xiang, et al., Analytical and clinical validity of whole-genome oligonucleotide array comparative genomic hybridization for pediatric patients with mental retardation and developmental delay, Am. J. Med. Genet. 146A (15) (2008) 1942–1954.
[9] C. Li, L. Liu, V. Dinu, Pathways of topological rank analysis (PoTRA): a novel method to detect pathways involved in hepatocellular carcinoma, PeerJ 6 (2018) e4571.
[10] C. Li, V. Dinu, miR2Pathway: a novel analytical method to discover MicroRNA-mediated dysregulated pathways involved in hepatocellular carcinoma, J. Biomed. Inform. 81 (2018) 31–40.
[11] S.E. Day, et al., Next-generation sequencing methylation profiling of subjects with obesity identifies novel gene changes, Clin. Epigenetics 8 (2016) 77.
[12] M.J. Huentelman, et al., Association of SNPs in EGR3 and ARC with Schizophrenia supports a biological pathway for schizophrenia risk, PLoS ONE 10 (10) (2015), e0135076.
[13] B.S. Bradley, et al., Differential expression of microRNAs as predictors of glioblastoma phenotypes, BMC Bioinformatics 15 (1) (2014) 21.
[14] A.L. Gallitano, et al., Family-based association study of early growth response gene 3 with child bipolar I disorder, J. Affect. Disord. 138 (3) (2012) 387–396.
[15] N. Briones, V. Dinu, Data mining of high density genomic variant data for prediction of Alzheimer's disease risk, BMC Med. Genet. 13 (2012) 7.
[16] V. Dinu, P.L. Miller, H. Zhao, Evidence for association between multiple complement pathway genes and AMD, Genet. Epidemiol. 31 (3) (2007) 224–237.
[17] K.S. Durgesh, B. Lekha, Data classification using support vector machine, J. Theor. Appl. Inf. Technol. 12 (1) (2010) 1–7.
[18] W.S. Noble, What is a support vector machine? Nat. Biotechnol. 24 (12) (2006) 1565–1567.
[19] N. Gupta, Artificial neural network, Network and Complex Syst. 3 (1) (2013) 24–28.
[20] Reese, H., Understanding the differences between AI, machine learning, and deep learning. 2017.
[21] Marée, R., et al. Decision trees and random subwindows for object recognition.
[22] Chen, T. and C. Guestrin. Xgboost: a scalable tree boosting system.
[23] Sklearn Page. Available online: https://scikit-learn.org/stable/supervised_learning.html (accessed on 23 May 2022).
[24] S.J. MacEachern, N.D. Forkert, Machine learning for precision medicine, Genome 64 (4) (2021) 416–425.